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Replacing a Tertiary-Connected 30 kV/50 MVAr Shunt Reactor With a STATCOM or SVC

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Abstract: Tertiary-connected shunt reactors absorb the capacitive surplus of lightly loaded extra-high-voltage (EHV) networks at least cost, but the absorbed megavars circulate through the host transformer as real winding current. This paper asks whether replacing such a reactor with a power-electronic compensator pays, and finds the answer to depend on the transformer's measured condition rather than on the technology. A 30 kV, 50 MVAr gapped-core reactor energised 80% of the year on the tertiary of a 280 MVA, 400/150/30 kV oil-forced air-forced (OFAF) autotransformer is compared against a ±50 MVAr modular multilevel converter (MMC)-based static synchronous compensator (STATCOM) at the tertiary, a 50 MVAr thyristor-controlled reactor (TCR)-based static var compensator (SVC) at the tertiary, and a stand-alone ±50 MVAr STATCOM at 150 kV. A statistical loss model shows the expected losses of a continuously controlled compensator depend on the duty only through its mean-square value; annual compensation losses fall from 2.22 GWh to 0.62–0.80 GWh (€142–160 k/yr). Insulation ageing is evaluated by hourly integration of the International Electrotechnical Commission (IEC) 60076-7 loss-of-life integral over 8,760 h of modelled ambient temperature, through-load and reactive duty. It confirms the relative benefits—the tertiary hot-spot falls 22 K on an annual mean and the unit ages 2.6–2.8 times more slowly—but shows the absolute rates in the design thermal state to be far below unity ageing, so bulk thermal duty cannot explain the degradation measured on a fleet unit. Diagnostics of that unit, whose furan sample post-dates oil reclamation, imply at least 0.70 equivalent years per year, reproduced only by a localised fault zone running at least 28 K above the bulk tertiary hot-spot; removing the tertiary current lowers that zone by at least a factor of 20 whatever the fault mechanism. Replacement is uneconomic against a healthy reactor and a thermally normal transformer, the SVC alone is positive on the diagnosed unit, both tertiary-connected options are positive on an end-of-life reactor, and where both conditions coincide every option is positive, at +€1.79 M to +€3.25 M.

Keywords: Shunt reactor; STATCOM; Static var compensator; Transformer insulation ageing; Reactive power compensation

1 Introduction

Shunt reactors remain the most compact and least expensive absorbers of reactive power on extra-high-voltage (EHV) transmission systems, and connection to the tertiary winding of a large network transformer is historically the cheapest installation option because it avoids a dedicated EHV bay [1]. The penalty is equally well documented: the absorbed reactive power circulates as real load current through the transformer windings, producing extra load losses inside the transformer, and the tertiary bus falls by roughly the transformer impedance drop whenever the reactor is energised [2].

In the Greek 400/150 kV system this penalty has materialised as an asset-management problem. In northern Greece, where long EHV corridors and a growing cable share produce a persistent capacitive surplus, tertiary reactors are typically energised about 80% of the year. Operating experience includes network transformers that failed in service or aged unexpectedly fast, with the damage concentrated in the tertiary winding insulation that carries the reactor current. The pattern is consistent with the international experience codified in IEEE Std C57.158, which identifies loaded tertiaries as a recurrent vulnerability of large three-winding transformers and autotransformers and notes that, with static synchronous compensators (STATCOMs) available, reactive compensation through transformer tertiaries has become less attractive [2].

The engineering decision examined here is whether to retain a fixed 50 MVAr shunt reactor on the tertiary of a 280 MVA autotransformer or to replace it with a continuously controlled compensator. Three alternatives are compared: a static var compensator (SVC) and a modular multilevel converter (MMC)-based STATCOM at the tertiary, and a stand-alone STATCOM at 150 kV. What is missing from earlier work is an assessment that carries measured transformer diagnostics through a standards-based thermal model into an investment decision. Published treatments evaluate the ageing law at one or two design points, and assert rather than quantify the transformer-protection benefit. This paper supplies that link. The transformer-protection argument is found to fail as a general claim but to hold as a conditional one, so the economic verdict follows the measured condition of the transformer rather than the technology.

The objectives are: (i) to derive the expected-loss saving of a continuously controlled compensator from the statistics of the reactive duty rather than from its rating; (ii) to integrate the IEC 60076-7 ageing law [3] hour by hour in place of the customary two-state approximation; (iii) to calibrate a localised fault zone against measured furan-derived life consumption rather than an assumed fault temperature; (iv) to quantify the harmonic burden a thyristor-controlled reactor (TCR) places on the tertiary; and (v) to translate the result into a three-scenario net-present-value comparison.

1.1 Literature Review

Two industry developments motivate the replacement options studied here. First, static compensation has matured. Thyristor-based SVCs have five decades of service history and a well-characterised loss and harmonic signature [4, 5]; the underlying converter concept of a controllable synchronous voltage source was set out by Gyugyi [6]. Voltage-sourced-converter STATCOMs built from modular multilevel cascade converter topologies are now standard utility equipment, offering continuously variable output in both signs, near-sinusoidal voltage synthesis without harmonic filters, and high partial-load efficiency [5, 7, 8]. The classification of the cascade family into single-star and single-delta bridge-cell and double-star chopper-cell and bridge-cell configurations [7] is the basis of the topology comparisons in studys [9] and [10], which address negative-sequence capability and semiconductor utilisation respectively; converter reliability and the influence of modulation strategy on device life are treated in study [11].

Second, the loading guides give a quantitative, standardised link from winding temperature to insulation life. IEC 60076-7 [3] provides the steady-state and differential thermal models, the cooling-mode exponents and the relative ageing rate for non-thermally-upgraded kraft paper; IEEE Std C57.91 [12] gives the Arrhenius equivalent for thermally upgraded paper. The chemical route to the same quantity—furanic compounds in oil correlated to the degree of polymerisation of the paper—is discussed by Cheim et al. [13], while de Pablo [14] specifically examined the relationship between furfural and insulation ageing. The chemical indicators of cellulosic insulation degradation are also reviewed by Emsley and Stevens [15]; CIGRE has since consolidated the interpretation of furanic markers [16] and the ageing kinetics of liquid-impregnated cellulose [17], and has documented field experience with solid-insulation ageing markers including the substantial spread between competing furan-to-DP correlations [18]. Dissolved-gas interpretation follows IEC 60599 [19].

On the specific question of transmission-level STATCOM connection, the relevant precedent is the 154 kV, ±50 MVAr transmission STATCOM implemented and field-tested in Türkiye [20]. It is important to state precisely what that installation demonstrates and what it does not. It comprises five 10.5 kV, ±12 MVAr cascaded multilevel converter modules operating in parallel and is connected to the 154 kV bus through a series filter reactor bank and a 154/10.5 kV coupling transformer; a companion study addresses saturation of that coupling transformer. It therefore establishes a working machine at the voltage and rating class considered here, but it is not a transformerless installation and does not constitute field precedent for direct connection. Transformerless cascaded and hybrid-multilevel designs for medium- and high-voltage service have been analysed by simulation [21], and transformerless Modular Multilevel Cascade Converter–Single-Delta Bridge-Cell (MMCC-SDBC) hardware has been verified experimentally at distribution voltage [22]. The stand-alone 150 kV option evaluated in Section 3.7 is accordingly treated as a feasibility case with an explicit commercial-maturity penalty, not as an off-the-shelf alternative.

Reactor-side experience completes the picture: switching duty and its transient consequences are covered by IEEE Std C37.015 [23], loss and sound-level determination by IEC 60076-6 [24], and failure investigation and repair-or-replace experience for a large shunt reactor population by Ryder et al. [25]. What is missing from this literature is an evaluation that carries measured transformer diagnostics through a standards-based thermal model into an investment decision. That is the gap addressed here.

2 Research Methodology

2.1 Case-Study System and Assumptions

The study object is a 280 MVA, 400/150/30 kV YNyn0d11 autotransformer with a 50 MVAr gapped-core oil-immersed shunt reactor breaker-switched on its 30 kV tertiary. Table 1 lists every parameter used anywhere in the

analysis, including the thermal and ageing quantities that were previously carried implicitly in the text. All quantities marked "assumed" are planning values, not manufacturer heat-run or tender data; the calculation is repeatable with project data by substituting them.

Table 1. Case-study system data, thermal parameters and study assumptions

Parameter	Value	Basis/Remark
Rated power HV/MV/TV	280/280/60 MVA	Nameplate
Rated voltages, vector group	400/150/30 kV, YNyn0d11	Nameplate
Impedances HV–MV/HV–TV/ MV–TV	13/38/24% on 280 MVA	Nameplate; star equivalent $Z_{\mathrm{H}} = 13.5\%$, $Z_{\mathrm{M}} = -0.5\%$, $Z_{\mathrm{T}} = 24.5\%$
No-load loss P_0	120 kW	Typical for the rating
Load loss HV–MV at 280 MVA	650 kW	Typical loss density
Load loss HV–TV pair at 60 MVA	300 kW (75 °C ref.)	$b_{\mathrm{T}} = 0.0833$ kW/MVAr^2
Loss ratio $R = P_{\mathrm{k}}/P_0$	5.4	Derived
Cooling; IEC 60076-7 exponents	OFAF; $x = 1.0$, $y = 1.3$; $H = 1.3$	IEC 60076-7 recommended values for OF cooling [3]
Rated top-oil rise $\Delta\theta_{\mathrm{or}}$	42 K	Typical OF-cooled value
Rated hot-spot gradients $H \cdot g_{\mathrm{r}}$, TV/MV	26 K/24 K	Specified by tender
Insulation	Non-thermally-upgraded kraft	98 °C unity-ageing reference, 6 K doubling [3]
Shunt reactor	50 MVAr, 30 kV, gapped core, breaker-switched	CIGRE TB 655 technology class [1]
Reactor losses (measured)	109 kW at 30 kV, 75 °C	Quality factor $q = Q/P = 459$; loss determination per IEC 60076-6 [24]
Reactor energisation	80% of the year (≈ 7,000 h)	Study brief, northern Greece
Ambient temperature	Hourly; annual mean 16.0 °C, seasonal amplitude 11 K, diurnal amplitude 5 K	Modelled for northern Greece (Section 2.5); January mean 5.0 °C, July mean 26.9 °C, 99th percentile 32.6 °C
Transformer through-load	Hourly; annual mean 0.55 p.u., peak 0.91 p.u.	Injection-point profile; sensitivity to 0.84 p.u. mean in Section 3.3
Insulation life budget	30 equivalent years (base); 17.12 eq-yr sensitivity	Planning value; IEC 60076-7 normal-life values are 17.12 and 20.55 eq-yr [3]. Section 3.5 shows the projection is budget-invariant
Furan-to-DP correlations; new/end-of-life DP	Chendong, Burton, de Pablo; $\mathrm{DP}_0 = 1{,}100$, $\mathrm{DP}_{\mathrm{end}} = 200$	[13–15, 18]
150 kV/400 kV short-circuit power	4,000/12,000 MVA	Moderate-strength buses
Wholesale energy price	€100/MWh central (€75–115 range)	Greek day-ahead market, 2024–2026 monthly averages [26]
Discount rate/horizon	7% real/25 years	Annuity factor 11.654

Note: HV = high voltage; MV = medium voltage; TV = tertiary voltage; OFAF = oil-forced air-forced; DP = degree of polymerisation.

Three classes of input are used, and the final column of Table 1 identifies each. Nameplate data cover the ratings, voltages, vector group and impedances. Measured data cover the reactor losses of 109 kW and the dissolved-gas, furan and dielectric results of Section 3.4. Everything else is an assumed planning value: the loss split, the rated thermal rises and gradients, the ambient and through-load series, the reactive duty, and the capital costs. The ranking of the alternatives is not sensitive to the assumed values. The loss ordering is set by the standing-loss floors and survives a threefold variation in the assumed duty (Section 3.2), and the investment ranking is governed by the

measured condition of the transformer. The capital costs are the only assumptions able to reorder the options.

2.2 Reactive-Power Duty

The reactor absorbs the charging power of the surrounding 400 kV network. The study assumes an absorption-duration curve with a maximum of 26 MVAr, mean 10 MVAr and annual root-mean-square value $Q_{\text{rms}} = 12.0$ MVAr, generated as an hourly series anti-correlated with the transformer through-load so that light-load hours carry the largest capacitive surplus. The reactor is taken to be energised during the 80% of hours with the lowest through-load, which reproduces the observed duty ratio and, more importantly, enforces the physical coupling between reactor duty and light network loading that a single-point analysis conceals. Two properties of this duty matter analytically. First, the requirement almost never approaches the 50 MVAr block size—reactor ratings follow standard sizes, not duty statistics—so the fixed reactor over-compensates heavily for some 7,000 h/yr. Second, the loss saving of a controllable device depends on the duty only through its mean-square value, as shown next.

2.3 Candidate Compensators, All Rated 50 MVAr Net Inductive

STATCOM at the tertiary. A ±50 MVAr MMC STATCOM in single-delta bridge-cell or double-star configuration [7–10], connected at 30 kV through phase reactors, reusing the existing reactor bay, breaker and protection; no coupling transformer is required at this voltage class. Published loss figures for MMC STATCOMs are of the order of 1% of rated apparent power at full output, falling steeply at partial output because conduction losses dominate [5, 7]. A value of 0.9% of 50 MVA (450 kW) at rated output is adopted, with a quadratic-plus-fixed characteristic and a 40 kW no-load floor covering auxiliaries, cooling and control.

SVC at the tertiary. A TCR with 5th- and 7th-harmonic filter branches at 30 kV. Because the filters contribute approximately 12 MVAr of permanently connected capacitive reactive power, the TCR must be rated 62 MVAr for the assembly to deliver 50 MVAr net inductive; the net control range is therefore -12 MVAr capacitive to +50 MVAr inductive. Losses are taken as 60 kW at zero net output—the point at which the TCR conducts 12 MVAr to cancel the filters—rising to 400 kW (0.8% of 50 MVA) at 50 MVAr net inductive, consistent with the loss breakdown given for SVC transformers, TCR branches, filters and auxiliaries in the FACTS literature [5].

Stand-alone STATCOM at 150 kV. A direct-connected machine with no coupling transformer, in which cascaded submodule strings withstand and synthesise the full system voltage, coupled through air-core phase reactors. The phase voltage of 86.6 kV rms implies roughly five times the submodule count of the 30 kV machine—of the order of 90–120 full-bridge cells per phase at 1.5–1.8 kV average cell voltage with redundancy—together with transmission-class insulation coordination and a 170 kV switchbay. Losses of 1.0% (500 kW) at rated output with a 45 kW no-load floor and no transformer loss component are adopted. As set out in Section 1.1, this configuration has no field precedent at transmission voltage and is evaluated as a feasibility case.

2.4 Statistical Loss Model

All four options have loss characteristics of the quadratic form:

$$P(Q) = a + b \cdot Q^2 \tag{1}$$

a being the standing loss (no-load converter losses, filter losses, or—for the fixed reactor while energised—the full constant block) and b collecting the quadratic conduction terms: converter or TCR conduction, and for tertiary-connected devices the transformer's HV–TV load-loss term $b_T = P_{k(\text{HV}-\text{TV})}/S_{\text{TV}}^2 = 300\,\text{kW}/(60\,\text{MVA})^2$. Taking the expectation of Eq. (1) over the annual duty $Q(t)$:

$$\text{E}[P] = a + b \cdot \mathbb{E}[Q^2] = a + b \cdot (\mu^2 + \sigma^2) = a + b \cdot Q_{\text{rms}}^2 \tag{2}$$

that is, the expected losses of any continuously controlled compensator depend on the duty only through its mean-square value; the mean μ and the variability σ enter only in that combination. This is the analytical core of the loss case: with $Q_{\text{rms}} = 12$ MVAr against a 50 MVAr rating, the quadratic terms are suppressed by $(12/50)^2 = 5.8\%$, so controllable devices operate almost at their standing-loss floor. The fixed reactor, by contrast, dissipates its full block losses throughout its 80% duty:

$$\text{E}[P_{\text{reactor}}] = d \cdot [P_R + b_T \cdot Q_R^2] = 0.80 \cdot [109 + 208]\,\text{kW} = 254\,\text{kW} \tag{3}$$

where, d is the duty ratio, $P_R = 109$ kW the reactor's measured losses and $Q_R = 50$ MVAr its block size. The 208 kW term is the transformer HV–TV load loss at $I_{\text{TV}} = 50/(\sqrt{3} \cdot 0.03) = 962$ A = 0.83 p.u. of the 60 MVA tertiary; the reactor's own 109 kW is dissipated in its own tank and does not enter the transformer thermal model. Eqs. (2) and (3) expose the design sensitivity: reactor losses scale linearly with duty d, while controllable-device losses scale with Q_{rms}^2.

2.5 Hourly Thermal and Loss-of-Life Model

For the oil-forced air-forced (OFAF)-cooled unit, IEC 60076-7 gives the steady-state hot-spot temperature as:

$$\theta_{\mathrm{HS}} = \theta_{\mathrm{a}} + \Delta\theta_{\mathrm{or}} \cdot \left[\frac{1 + RK^2}{1 + R}\right]^{x} + H \cdot g_{\mathrm{r}} \cdot K^{y} \tag{4}$$

with ambient θ_{a}, rated top-oil rise $\Delta\theta_{\mathrm{or}}$, loss ratio R, load factor K, hot-spot factor H, rated winding gradient g_{r}, and the recommended OF-cooling exponents $x = 1.0$ and $y = 1.3$ [3]. Two OFAF-specific consequences follow. With $x = 1.0$ the top-oil rise is linear in total losses, so removing a given block of compensation loss buys a proportionally larger oil-temperature reduction than in an ON-cooled unit, where $x \approx 0.8$ flattens the response. With $y = 1.3$ the winding gradient is less steeply current-dependent than the ON value of 1.6, which moderates the gradient saving; the two effects partially offset and both are captured by the integration below.

Because $x = 1.0$, the load-factor bracket in Eq. (4) is evaluated directly as a loss ratio, the reactor or compensator contribution entering the numerator as an additional load-loss block:

$$\Delta\theta_{\mathrm{o}}(t) = \Delta\theta_{\mathrm{or}} \cdot \frac{P_0 + P_k K(t)^2 + b_T Q(t)^2}{P_0 + P_k} \tag{5}$$

IEC 60076-7 defines the relative ageing rate of non-upgraded kraft paper as $V(\theta_{\mathrm{HS}}) = 2^{(\theta_{\mathrm{HS}} - 98)/6}$, a doubling per 6 K [3], IEEE Std C57.91 giving the Arrhenius equivalent with a 110 °C reference for upgraded paper [12]. The insulation loss of life over a period T is the integral:

$$L = \int_0^T V(\theta_{\mathrm{HS}}(t))\, dt \tag{6}$$

expressed in equivalent ageing years against a notional life budget. Previous treatments of this problem, including the author's own, evaluate Eq. (6) at one or two representative states. That approximation is unsafe here, because V is exponential in temperature while ambient temperature, transformer loading and reactive duty are all strongly and mutually correlated in time. Eq. (6) is therefore integrated over 8,760 hourly steps. The ambient series is synthesised as a seasonal sinusoid plus a diurnal sinusoid plus Gaussian residual, calibrated to northern Greek climatology (Table 1). The through-load series combines a winter-peaking seasonal term, a summer air-conditioning term and a double daily peak, scaled to an annual mean of 0.55 p.u. The reactive duty is the series of Section 2.2. Tertiary and MV winding hot-spots are evaluated from Eqs. (4)–(5) at each hour and V accumulated. The three input and output layers of this integration are presented in Section 3.3.

The same integration is used to calibrate the localised fault of the diagnosed unit. Rather than assuming a fault temperature, a constant offset $\Delta\theta_{\mathrm{f}}$ is added to the bulk tertiary hot-spot during reactor-on hours and solved so that the modelled annual rate reproduces the furan-derived measured rate. The offset is then held fixed, or allowed to grow linearly, to project forward.

2.6 Furan-Derived Life Consumption: Stating the Convention

Furan-to-DP correlations return a degree of polymerisation; converting that DP to a fraction of insulation life consumed requires a further and rarely stated choice of scale. Three conventions are in use: linear in DP, linear in the reciprocal of DP (the chain-scission scale that follows from first-order depolymerisation kinetics [15, 17]), and linear in $\log \mathrm{DP}$. For DP_0 = 1,100 and $\mathrm{DP}_{\mathrm{end}}$ = 200 they differ by a factor of 2.6 at DP = 515. This paper adopts the logarithmic scale and states so explicitly; Section 3.5 reports the projection under all three, because the choice propagates directly into the remaining-life estimate and hence into the economics. Failing to specify the convention is, in the author's view, the single most common source of non-comparability between published furan-based life assessments.

2.7 Harmonic Model

The TCR line current is constructed numerically for firing angles spanning the full control band and decomposed by fast Fourier transform, avoiding closed-form approximations. Because the TCR is delta-connected, triplen harmonics circulate within the delta and are absent from the current entering the transformer tertiary—a topological property, not a filtering effect, and one that no line-side filter could reproduce. The remaining characteristic orders are weighted according to IEEE Std C57.110 [27], under which the winding eddy-loss component scales as $\sum_h I_h^2 h^2$, to give the harmonic loss factor:

$$F_{\mathrm{HL}} = \frac{\sum_h I_h^2 h^2}{\sum_h I_h^2} \tag{7}$$

Two filter-performance cases are evaluated: filters in tune, attenuating the 5th and 7th orders to 10% of the injected current and the 11th and higher orders to 40%; and filters detuned by component ageing, at 30% and 60% respectively. Critically, F_{HL} is a ratio and says nothing about absolute heating, so the eddy-loss increment is also expressed in kilowatts against the tertiary load loss at the prevailing current.

2.8 Cost–Benefit Framework

Greek day-ahead prices ranged between roughly €74/MWh and €112/MWh in monthly averages through 2025, after the 2022 crisis peak above €400/MWh [26]. Transmission losses are settled at wholesale-related prices, so the day-ahead level is the appropriate marginal valuation; €100/MWh is used centrally with €75–115/MWh sensitivity. All flows are discounted at 7% real over 25 years, giving an annuity factor of 11.654. Indicative turnkey capital costs of €3.0 M (tertiary SVC), €5.0 M (tertiary STATCOM, bay reused) and €6.5 M (150 kV machine including bay) are used. These figures were selected as order-of-magnitude planning values. They scale with the installed power-electronic content and with the switchbay work each option requires: the tertiary SVC reuses the existing 30 kV bay, the tertiary STATCOM adds converter content at the same voltage, and the 150 kV machine adds roughly five times the submodule count together with a 170 kV bay. Their relative spacing follows the cost ranking reported for these technology classes in the FACTS literature [5]. No published price series for this equipment class is sufficiently specific or archival to be cited. They are therefore not quotations. An actual project decision would require current supplier quotations and site-specific installation and civil costs, which may reorder the three options. Procurement of a STATCOM against IEEE Std 1052 [28] should in any case require type tests and service references, which for the direct-connected option would be the binding constraint. Three scenarios are evaluated:

Scenario A—base: healthy reactor, transformer in its design thermal state. Benefits are energy-loss savings, the transformer-life value of the modelled ageing slowdown, and avoided reactor mid-life refurbishment.

Scenario B—end-of-life reactor with dynamic support valued: the reactor requires replacement in any case, an outcome documented in utility failure-investigation experience [25]. An avoided new reactor (€1.2 M) is credited, together with a dynamic-support value of €5/kVAr·yr on the ±50 MVAr range (€250 k/yr), of which 40% is credited to the slower, voltage-squared-limited SVC. That unit value is an explicit planning assumption and should be replaced by the transmission system operator's own reactive-support valuation. In this scenario the residual value of the forgone healthy reactor is removed from the balance, since the reactor is no longer healthy.

Scenario C—diagnosed degraded tertiary: healthy reactor, but the transformer is in the measured condition of Section 3.4. The transformer-protection value derived in Section 3.5 replaces Scenario A's design-state life value.

3 Results and Discussion

3.1 Compensation Losses

Figure 1 plots the Eq. (1) characteristics, with the transformer-internal component included for tertiary-connected devices and the two-state reactor shown as a level rather than a curve, and Figure 2 the assumed duty-duration curve with the reactor's 0/50 MVAr block at 80% energisation. At the full 50 MVAr point the passive reactor is the most efficient device—317 kW against 500–658 kW for the converters. Power electronics do not win at the rated point; they win because the duty almost never visits it and Eq. (2) collapses their expected losses toward the standing floor. Applying Eq. (2) with $Q_{\mathrm{rms}} = 12.0$ MVAr and the Greek day-ahead price level [26] gives Table 2.

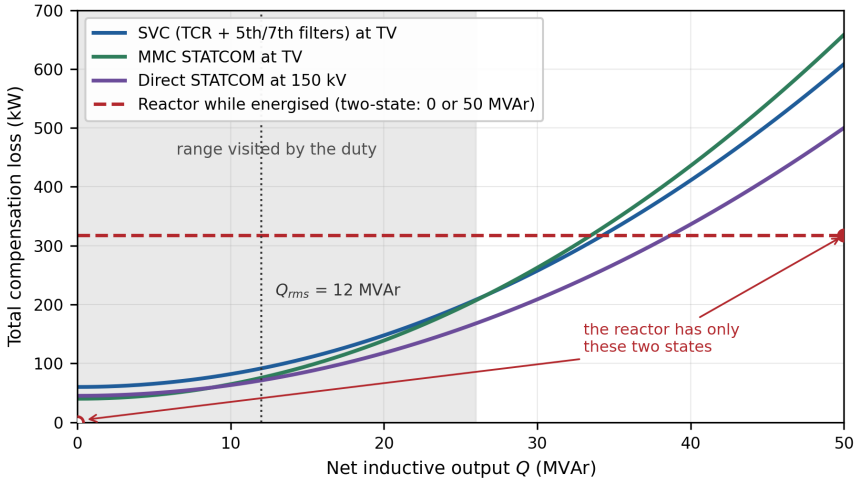


Figure 1. Total compensation losses versus net inductive output

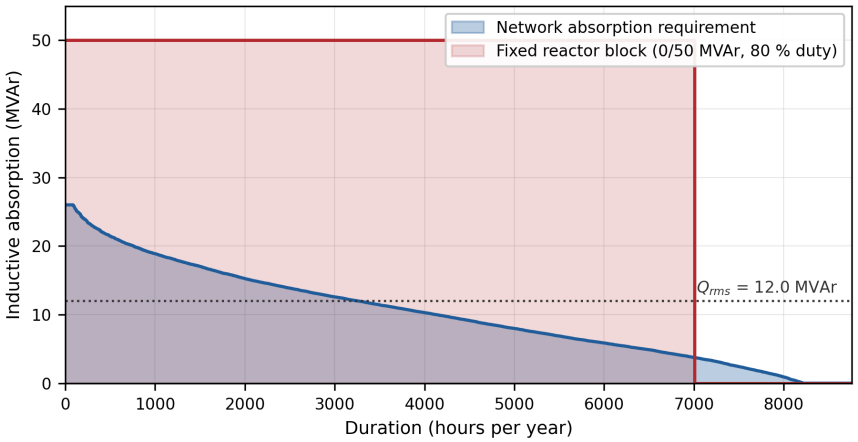


Figure 2. Assumed annual reactive-absorption duration curve

Table 2. Annual compensation losses, energy, and cost

Option	Average Losses (Eqs. (2)–(3))	Annual Energy	Annual Cost at €100/MWh	Saving vs Baseline
Reactor at TV (baseline, 80% duty)	254 kW	2.22 GWh	€222 k	—
SVC (TCR + filters) at TV	92 kW	0.80 GWh	€80 k	1.42 GWh/yr
MMC STATCOM ± 50 MVAr at TV	76 kW	0.66 GWh	€66 k	1.56 GWh/yr
Direct STATCOM ± 50 MVAr at 150 kV	71 kW	0.62 GWh	€62 k	1.60 GWh/yr

Note: TV = tertiary voltage; SVC = static var compensator; TCR = thyristor-controlled reactor; MMC = modular multilevel converter; STATCOM = static synchronous compensator. "—" indicates not applicable.

3.2 Sensitivity of the Loss Case to the Assumed Duty

Q_{rms} is the single input on which the entire loss case rests, and it is assumed rather than logged. Table 3 therefore reports the loss result for the duty series scaled, with its shape retained, between half and 1.5 times its base amplitude. The saving is largely insensitive: doubling Qrms from 6 to 12 MVAr costs only €0.27 M of net present value on the tertiary STATCOM, and even at $Q_{\mathrm{rms}} = 18$ MVAr—half the reactor block size—all three options still save more than 1 GWh/yr. This follows directly from the quadratic suppression in Eq. (2), and is among the more robust results reported here. It also means that logging the actual reactive duty, while desirable, is not a prerequisite for the loss decision.

Table 3. Sensitivity of annual losses and energy-saving net present value to the assumed reactive duty

Q_{rms} (MVAr)	SVC (kW)	STATCOM TV (kW)	STATCOM 150 kV (kW)	SVC Saving NPV (M€)	STATCOM TV NPV (M€)	STATCOM 150 kV NPV (M€)
6.0	68	49	52	+1.90	+2.09	+2.07
9.0	78	60	60	+1.80	+1.98	+1.98
12.0 (base)	92	76	71	+1.66	+1.82	+1.86
15.0	109	96	86	+1.48	+1.62	+1.71
18.0	131	120	104	+1.25	+1.37	+1.53

Note: Q_{rms} = root-mean-square reactive power; SVC = static var compensator; STATCOM = static synchronous compensator; TV = tertiary voltage; NPV = net present value.

3.3 Hourly Thermal and Ageing Results

Table 4 and Figure 3 give the outcome of integrating Eq. (6) hour by hour. Figure 3 shows the modelled ambient temperature, the tertiary hot-spot under reactor duty and after replacement, and the relative ageing rate on a logarithmic scale, including the calibrated fault-affected paper of the diagnosed unit. Three findings must be separated, because they point in different directions and the paper's earlier single-point treatment conflated them.

Table 4. Hourly-integrated thermal and ageing results

Quantity	Reactor (Baseline)	SVC/ STATCOM at TV	Direct STATCOM at 150 kV	High-Utilisation Bounding Case (Reactor)
Top-oil rise, annual mean	26.8 K	18.3 K	17.7 K	41.5 K
Tertiary hot-spot, annual mean/max	59/88 °C	37/57 °C	34/56 °C	74/108 °C
MV winding hot-spot, annual mean/max	54/81 °C	45/71 °C	45/71 °C	—/111 °C
Tertiary loss-of-life rate (Eq. (6))	0.027 eq-yr/yr	0.0014 eq-yr/yr	0.0009 eq-yr/yr	0.149 eq-yr/yr
Whole-unit (MV) loss-of-life rate	0.0097 eq-yr/yr	0.0037 eq-yr/yr	0.0035 eq-yr/yr	0.191 eq-yr/yr
Whole-unit relative ageing	1× (reference)	2.6× slower	2.8× slower, including events	—

Note: Eqs. (4)–(6) integrated over 8,760 h. Hot-spot temperatures are rounded to 1 °C; the unrounded annual-mean MV-winding reduction equals the top-oil reduction of 8.4 K. MV = medium-voltage winding; TV = tertiary voltage; SVC = static var compensator; STATCOM = static synchronous compensator. "—" indicates not applicable.

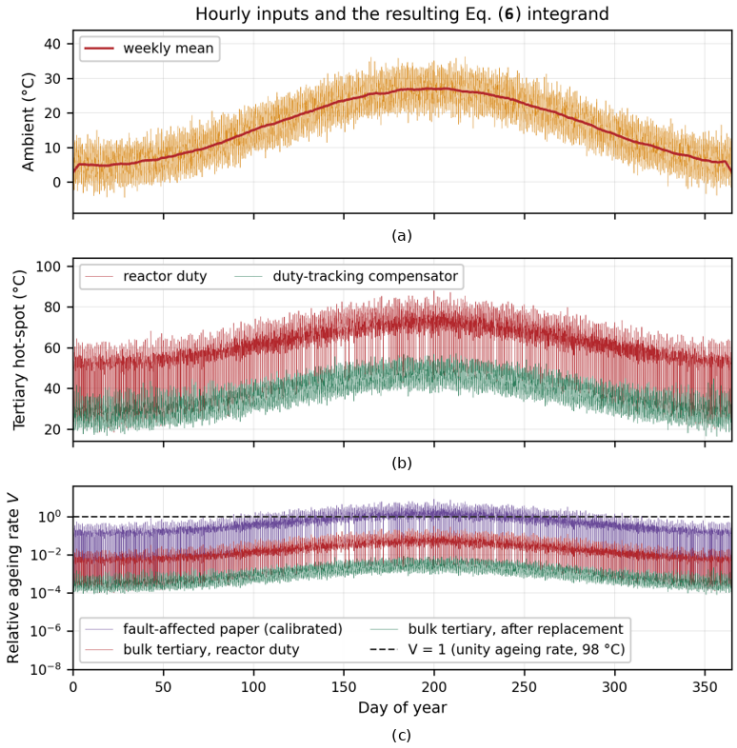


Figure 3. Hourly inputs and Eq. (6) integrand over one year: (a) ambient temperature; (b) tertiary hot-spot; (c) relative ageing rate

The relative benefits are confirmed. Removing the 208 kW reactor-driven load loss lowers the top-oil rise by 8.4 K on an annual-mean basis and 10.6 K during the hours when the reactor is actually energised. The tertiary hot-spot falls by 22.1 K on an annual mean and 27.7 K during reactor-on hours, and every other winding—including the limiting MV winding—runs 8.4 K cooler on an annual mean, the full top-oil reduction, because its gradient is set by the through-load alone; at its annual maximum, which falls in a reactor-on hour, the MV winding runs 10 K cooler. The whole-unit relative ageing rate, taken at the MV winding, falls by a factor of 2.6 for the tertiary-connected options and 2.8 for the 150 kV machine. These figures are close to the 25 K and 2.5-fold values obtained from a single summer design point, so the relative thermal case survives the more rigorous treatment intact.

The absolute rates do not support the design-state economic case. In the design thermal state the tertiary ages at 0.027 equivalent years per year under reactor duty and 0.001 after replacement; the MV winding ages at 0.010 and 0.004 respectively. Figure 4 places these bulk operating points on the IEC 60076-7 ageing curve, together with the calibrated fault-affected paper of Section 3.5. All four are one to three orders of magnitude below unity

ageing, because the hours of heavy reactor duty are by construction the hours of light network loading and, in a Mediterranean climate, are heavily weighted toward cool months. A 2.6-fold reduction in a rate of 0.010 eq-yr/yr saves approximately 0.15 equivalent years over a 25-year horizon, which on a €9 M asset with a 30-year budget is worth under €0.05 M in present value. The design-state transformer-life benefit of replacement is, in short, real but negligible. This directly contradicts the author's earlier estimate of €0.9 M, which rested on evaluating the ageing law at a summer peak-load condition that does not coincide with reactor duty. The engineering meaning of Table 4 is therefore that its two halves are not in conflict. Replacement delivers a large proportional gain, the tertiary running 22 K cooler and the unit ageing 2.6–2.8 times more slowly. That gain is applied, however, to an ageing rate already two orders of magnitude below the rate the insulation is designed to tolerate, so the life actually saved is small. A large relative improvement in a quantity that is not limiting does not create economic value. For a transformer in a normal thermal condition the case for replacement must therefore rest on losses and dynamic performance, not on insulation life.

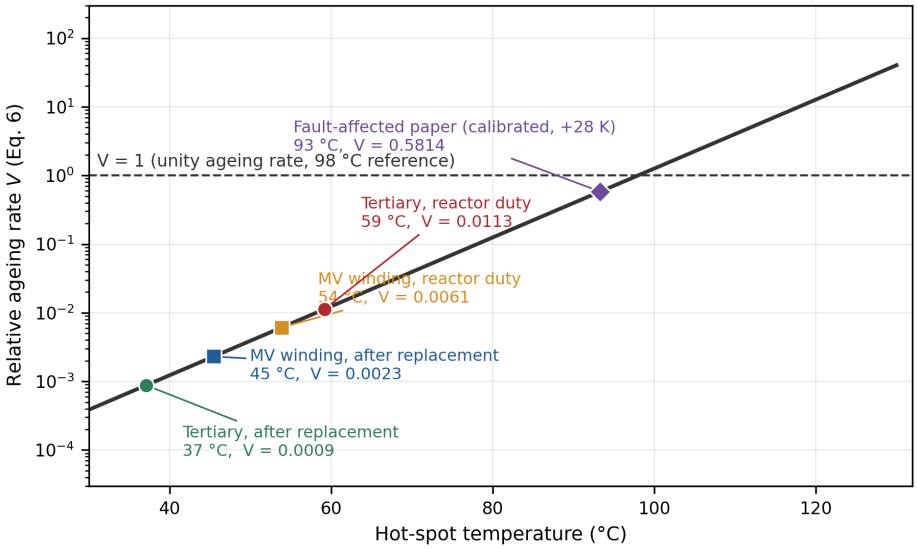


Figure 4. IEC 60076-7 relative ageing rate with the study operating points

The high-utilisation bounding case restores the earlier picture only under loading that contradicts the premise. Repeating the integration with the through-load scaled to an annual mean of 0.84 p.u. raises the tertiary hot-spot to a mean of 74 °C with a peak of 108 °C and the design-state tertiary ageing rate to 0.149 eq-yr/yr, with the MV winding peaking at 111 °C. That is the regime in which a single-point design calculation implicitly sits. It is internally inconsistent with the 80% reactor duty, however, since a transformer loaded to 0.84 p.u. on annual average does not sit on a network with a persistent capacitive surplus for 7,000 h a year. The two cases bracket the plausible range; the base case is used throughout, and the bounding case is reported so that the reader can rescale the design-state life benefit for a more heavily loaded installation.

3.4 Measured Diagnostics of a Degraded Fleet Unit

The design-state calculation assumes the tertiary behaves as the nameplate model says it should. The measured condition of a 280 MVA-class, 400/150/30 kV unit of the studied fleet, commissioned in 2007 with diagnostics spanning 2007–2026, shows what sustained reactor duty is associated with in practice. Table 5 summarises the record. Three caveats are stated at the outset, because each limits the inference that can be drawn.

3.5 Calibrated Fault Zone, Remaining Life and the Value of Averting Premature Replacement

On the logarithmic scale the Chendong DP of 515 corresponds to 44.5% of the insulation life budget consumed, or 13.4 equivalent years of 30, over the 19 years since commissioning—a measured average of 0.70 eq-yr/yr. The furan sample was taken after the 2025–2026 reclamation, which strips furans from the oil, so 0.51 ppm understates the equilibrium concentration: the DP of 515 is an upper bound, and the 44.5% consumption and the 0.70 eq-yr/yr rate are lower bounds. The hourly model of Section 3.3 produces 0.027 eq-yr/yr for the bulk tertiary in the same period. The measured rate is therefore not 1.7 times the modelled rate, as a single-point calculation suggested, but at least 26 times it. That discrepancy is the central diagnostic result of this paper: sustained tertiary loading, on its own and in this thermal state, cannot produce the observed degradation. The degradation is governed by the localised IEC 60599 T1 fault, and the reactor duty is at most a contributing thermal bias.

Table 5. Measured diagnostics of the degraded fleet unit

Diagnostic	Measured Values	Interpretation and Limitations
PI, main-to-tertiary insulation (H+L–Tert), 2010 and 2014	0.99–1.00, against 2.70–2.86 for windings-to-earth	Anomalous loss of dielectric absorption in the main–tertiary structure. Absolute insulation resistance remains in the gigaohm range, and PI loses diagnostic meaning at very high insulation resistance because the absorption current becomes negligible relative to the geometric charging current of what is a small insulation structure. Treated here as an anomaly warranting follow-up, not as proof of dielectric exhaustion; measurement voltage and temperature correction should be recorded before it is relied upon
Dissipation factor of CT, 2007 $\rightarrow$ 2014	0.59–0.67% $\rightarrow$ 0.78%	Slowly rising dielectric losses, within limits but trending
Dissolved gases, 2014 $\rightarrow$ 2026 (ppm)	CH_4 725–1,116; C_2H_6 592–1,028; C_2H_4 173–419; $C_2H_2 \leq 1$; CO 701–1,310; CO_2 7,023–18,940; H_2 621 (2014); TCG 2,245–3,043	IEC 60599 ratio coding of the 2014 sample gives CH_4/H_2 = 1.80 (>1), C_2H_4/C_2H_6 = 0.41 (<1) and $C_2H_2/C_2H_4 \approx 0$, i.e. fault type T1, a prolonged thermal fault below 300 °C [19]; C_2H_4/C_2H_6 stays between 0.29 and 0.41 in every later sample. The CO_2/CO ratio lies between 7.7 and 14.5 over 2014–2026, at 14.5 in 2014 and 13.2 in 2026; IEC 60599 associates paper involvement with carbonisation with ratios below 3, and ratios above 10 with mild paper overheating or oil oxidation. The carbon oxides therefore do not corroborate acute cellulose carbonisation. The furan data below carry the cellulose claim
Furan derivatives, 2026	2FAL 0.51 ppm (highest of the fleet); 5MEF 0.22 ppm	2FAL of 0.51 ppm lies in the moderate-ageing band; 5MEF presence is consistent with sustained paper overheating [13, 16]. Two limitations: furans are a bulk-oil measurement and cannot localise degradation to the tertiary, and the 2026 sample was taken after the 2025–2026 reclamation of the oil, which strips furans. 0.51 ppm is therefore a lower bound and every rate derived from it is conservative
Degree of polymerisation (from 2FAL)	DP 515 (Chendong)/558 (Burton)/756 (de Pablo)	On the logarithmic scale with $DP_0 = 1{,}100$ and $DP_{end} = 200$ this is 22–45% of life consumed, the spread arising from the correlations alone [13–15, 18]
Oil condition, 2014 $\rightarrow$ 2026	Breakdown voltage 54.7 $\rightarrow$ 92 kV; moisture 3–21 ppm; elevated particle counts (ISO 18/16/13)	Oil reclaimed 2025–2026, before the 2026 gas and furan samples; good dielectric strength; particle filter service indicated

Note: PI = polarisation index; CT = tertiary capacitance; H+L–Tert = high- and low-voltage windings to tertiary insulation; 2FAL = 2-furfuraldehyde; 5MEF = 5-methyl-2-furfural; DP = degree of polymerisation; TCG = total combustible gas; ISO = International Organization for Standardization. Assessment of record: tertiary condition severe-chronic.

Solving for the constant offset that reproduces the measured rate gives $\Delta\theta_f = 28.3$ K, a lower bound, placing the fault-affected paper at a mean of 93 °C during reactor-on hours and a peak of 116 °C. This is physically coherent: IEC 60599 notes that significant rates of polymer chain scission occur above 105 °C [19], which the calibrated fault zone exceeds during the warmer part of the year while the bulk winding never approaches it. It also explains why the failure mode escapes top-oil-based monitoring, the tertiary being the smallest and least instrumented winding.

The benefit of replacement does not depend on the fault being current-driven. This is the point on which the earlier version of this analysis over-claimed, and it can now be settled by calculation rather than assertion. Two bounding branches exist. If the fault is driven by the tertiary load current—through the current itself, its stray flux or its harmonics—removing that current eliminates the offset and the fault zone reverts to the bulk rate of 0.0014 eq-yr/yr. If instead the fault is a fixed-dissipation defect such as a poor joint or a circulating current in a clamp or shield, the offset persists, but the fault zone still cools with the bulk oil, and the rate falls to 0.036 eq-yr/yr. The improvement is therefore between 20-fold and 500-fold, and the lower bound is what is credited in the economics. The exponential form of Eq. (6) is what makes the conclusion robust to the mechanism: a 22 K reduction in the surrounding temperature is worth a factor of 12 in ageing rate regardless of what is producing the local hot spot.

The remaining-life projection is invariant to the life budget but not to the DP scale. Because the forward rate is calibrated against the same furan-derived consumption that defines the fraction already used, the absolute size of the life budget cancels: the same 23.7-year projection is obtained on a 30-equivalent-year budget and on the 17.12-equivalent-year normal-life value of IEC 60076-7. This removes what would otherwise be a significant

arbitrary choice. The DP-to-life scale does not cancel. Table 6 gives the projection under all three conventions and the resulting spread is an order of magnitude, from 10 to 56 years at the measured rate. Any furan-based remaining-life figure quoted without its scale is therefore uninformative.

Table 6. Remaining life under three DP-to-life-consumption conventions

Convention	Life Consumed, DP 515	Life Consumed, DP 756	Remaining Life at Measured Rate, DP 515	Remaining Life at Measured Rate, DP 756
Logarithmic in DP (adopted)	44.5%	22.0%	23.7 yr	67.4 yr
Chain scission, linear in $1/\mathrm{DP}$	25.2%	10.1%	56.3 yr	168.9 yr
Linear in DP	65.0%	38.2%	10.2 yr	30.7 yr

Note: Remaining life is evaluated at the measured ageing rate with DP_0 = 1,100 and $\mathrm{DP}_{\mathrm{end}}$ = 200; because the furan sample post-dates oil reclamation, the life-consumed values are lower bounds and the remaining-life values upper bounds. DP = degree of polymerisation.

Under the adopted convention and a constant fault offset the budget is exhausted in at most 24 years. Progression of the fault, which the gas record supports—the 2014 sample was already heavily gassed only seven years into service, and the total combustible gas content rose from 2,617 ppm in 2020 to 3,043 ppm in 2023, before reclamation reset it—shortens this sharply because the ageing law is exponential: an offset growing at 0.5, 1.0 and 1.5 K per year exhausts the budget in at most 15, 12 and 10 years respectively. A growth rate of 1.0 K/yr, giving 12 years, is adopted as the central fault-progression case. Figure 5 plots the trajectories from the measured 2026 state under continued reactor duty at the measured rate and under fault progression, and after replacement for both fault-mechanism branches. The reclamation bias has a known direction but an unknown size. As an indication, a pre-reclamation 2FAL of twice the measured value, 1.02 ppm, gives a Chendong DP of 429, 55.2% of life consumed and 0.87 eq-yr/yr, 32 times the modelled bulk rate; the constant-offset projection shortens to 15 years and the central progression case to 9 years. Every downstream quantity moves in the direction that strengthens the case for replacement, so the results reported at 0.51 ppm are conservative.

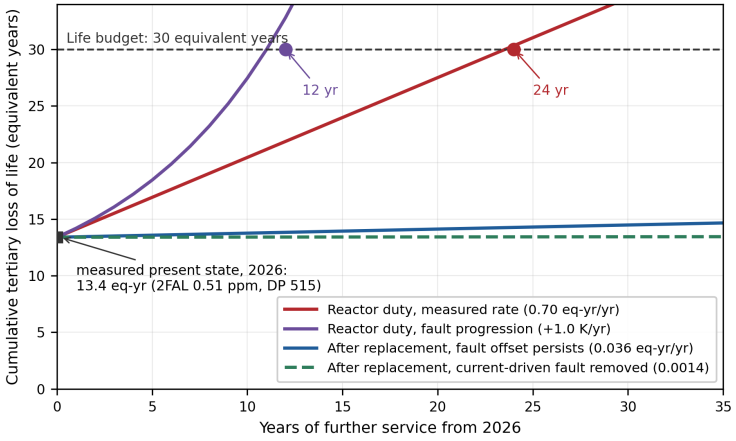


Figure 5. Cumulative tertiary insulation loss of life of the diagnosed unit

Capital value of averting premature replacement. If the fault-progression trajectory forces transformer replacement in 12 years instead of a scheduled retirement at 20 years—a 40-year life for the 2007 unit—the deferral value of avoiding the premature spend on a €9 M unit is, at 7% real:

$$\Delta\mathrm{NPV} = 9 \cdot (1.07^{-12} - 1.07^{-20}) = 9 \cdot (0.444 - 0.258) = €1.67\ \mathrm{M} \tag{8}$$

The corresponding figures for the 10- and 15-year progression cases are €2.25 M and €0.94 M. To this is added the risk-weighted cost of an in-service failure rather than a planned retirement—emergency procurement premium, extended unavailability of a 280 MVA injection point, collateral damage and site remediation—conservatively €3 M of consequence at a 30% probability that the end-of-life event is a failure rather than a condition-triggered retirement, giving €0.9 M expected. The transformer-protection value is therefore €2.57 M; €2.57 M is credited to the 150 kV option, which relieves the tertiary under all conditions, and €2.40 M to the tertiary-connected options, which remove the chronic duty but still pass full output through the tertiary during system events. Note that this value is roughly 30% below the figure obtained from the earlier single-point analysis, and that it is now conditional on

fault progression rather than on the measured rate, which by itself projects beyond the scheduled retirement date and generates no deferral benefit at all. Because the furan sample post-dates reclamation, these values are lower bounds: at twice the measured 2FAL the central-case deferral value rises from €1.67 M to €2.57 M.

3.6 Harmonic Burden: Quantified Rather Than Asserted

Winding eddy losses under non-sinusoidal current grow with the square of harmonic order, so a TCR on the tertiary appears at first sight to inject harmonic heating into the very winding the project is trying to protect. Figure 6 shows what the FFT decomposition and Eq. (7) actually give. Three points follow.

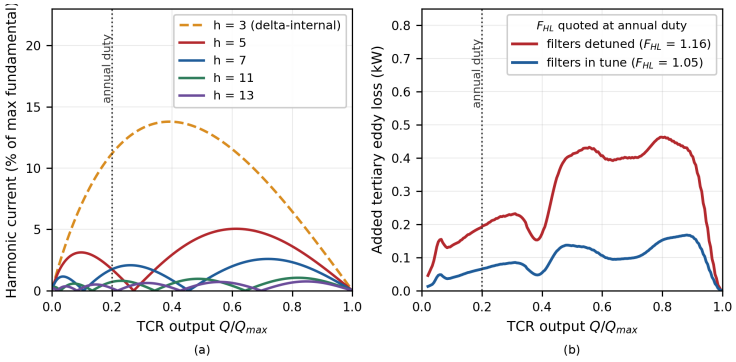


Figure 6. TCR harmonic currents (a) and added tertiary winding eddy loss (b) across the control band

Note: (a) Characteristic harmonic currents as a percentage of the maximum fundamental; the third harmonic circulates within the delta and does not enter the transformer tertiary. (b) Absolute increment in tertiary winding eddy loss, weighted per IEEE Std C57.110 over harmonic orders up to the 31st, for filters in tune and detuned, with the harmonic loss factors at the annual duty point quoted in the legend. TCR = thyristor-controlled reactor; h = harmonic order; F_{HL} = harmonic loss factor; $Q/Q_{\max}$ = ratio of reactive power output to maximum reactive power output.

First, the worst-case emission of every order occurs mid-band, not at the endpoints: the fifth harmonic peaks at 5.05% of the maximum fundamental at 61% of full output and the seventh at 2.59% at 72%, while at zero and full conduction the current is sinusoidal. A compensator sized for 50 MVAr but operating at an annual rms of 12 MVAr therefore sits permanently in the emitting region, which a rated-point harmonic study would miss entirely.

Second, the third harmonic is by far the largest characteristic order, reaching 13.78% of the maximum fundamental at 39% output, but it circulates within the delta-connected TCR and never reaches the transformer tertiary. It is a burden on the TCR branch itself, to be accounted for in the reactor and thyristor rating, and it is invisible to line-side filters by topology. Conflating the delta-internal spectrum with the tertiary-injected spectrum is a common source of overstated harmonic concern in tertiary SVC studies.

Third, the burden that does reach the tertiary is small. At the annual duty point the harmonic loss factor is 1.05 with filters in tune and 1.16 detuned; at the worst-case band point it is 1.01 and 1.03. Because the harmonic currents scale with a fundamental current that is itself only 0.17 p.u. of the tertiary rating at annual duty, the absolute increment in tertiary eddy loss is 0.07 kW in tune and 0.19 kW detuned, against a tertiary load loss of 8.3 kW at that current and 300 kW at rating. Across the whole control band the increment never exceeds 0.17 kW in tune or 0.46 kW detuned. The author's earlier characterisation of this as "not a secondary consideration" is not supported: it is a secondary consideration, by three orders of magnitude in absolute heating. The ranking of the alternatives must therefore rest on the thermal, availability and control arguments, and filter detuning is a maintenance rather than an asset-integrity issue. The MMC options retain a genuine but modest advantage in synthesising near-sinusoidal current without filters at all.

3.7 The Stand-Alone 150 kV Option

The cascaded-multilevel architecture removes the fundamental need for a coupling transformer: stacking enough full-bridge submodules per phase produces a converter that withstands and synthesises transmission-level voltage directly, as HVDC modular multilevel valves do. As established in Section 1.1, however, the 154 kV field precedent [20] is transformer-coupled, and the transformerless designs in the literature are either simulation studies at medium and high voltage [21] or hardware verified at distribution voltage [22]. No direct-connected machine at 150 kV and 50 MVAr is known to be in service. The option is therefore evaluated on its merits with the commercial risk stated, not presented as demonstrated practice.

Complete transformer relief. The tertiary reverts to a buried stabilising winding in the sense of IEEE Std C57.158 [2]: zero compensation-related gradient at all times, whereas the tertiary-connected options still pass full ±50 MVAr through the tertiary during system events; no external 30 kV fault or switching-surge exposure; no

stray-flux heating from tertiary load current; and no harmonic duty. For a unit in the measured condition of Section 3.4 this is the only option that removes the candidate damage driver categorically.

Availability decoupled from the transformer. A tertiary-connected compensator is lost whenever the 280 MVA unit is out—precisely when the network is weakened, and unavoidably during the extended electrical testing, including sweep-frequency response analysis and internal inspection, that the diagnosed unit's condition assessment recommends. The 150 kV machine keeps regulating through transformer outages and can be relocated in future network reconfigurations.

Direct leverage on the controlled bus. Vars injected at the tertiary traverse the star-equivalent tertiary reactance, X_{T}= 24.5% on 280 MVA , derived from the nameplate pair impedances. The converter terminal voltage moves by $X_{\mathrm{T}} \cdot \Delta Q/S$ = 4.4% per 50 MVAr, that is ±4.4% about nominal and 8.8% across the full ±50 MVAr range, consuming converter voltage margin and de-tuning the bus-voltage loop. The 150 kV machine acts directly at the controlled bus. For reference, switching the fixed reactor steps the 150 kV bus by 50/4,000 = 1.25% per operation, with the associated switching transients covered by IEEE Std C37.015 [23].

Costs. Roughly five times the power-electronic content, transmission-class insulation coordination and a 170 kV bay, together with a thinner commercial track record than transformer-coupled products. The indicative €6.5 M used in Section 3.8 carries more uncertainty than the other two capital figures for exactly this reason.

3.8 Cost–Benefit Results

Table 7 and Figure 7 give the three-scenario net present values, including the case in which the Scenario B and C conditions coincide.

Table 7. Three-scenario cost–benefit results

NPV Item Over 25 Years (M€)	SVC at TV	STATCOM at TV	Direct STATCOM 150 kV
Energy-loss savings (1.42/1.56/1.60 GWh·yr^{-1} at €100/MWh × 11.654)	+1.66	+1.82	+1.86
Transformer-life value, design state (Section 3.3)	+0.05	+0.05	+0.07
Avoided reactor mid-life refurbishment and maintenance	+0.20	+0.20	+0.20
Incremental O&M of the new asset	-0.37	-0.42	-0.46
Capital cost (indicative turnkey)	-3.00	-5.00	-6.50
Residual value of the healthy reactor forgone	-0.10	-0.10	-0.10
Net NPV—Scenario A (base)	-1.56	-3.45	-4.93
Net NPV—Scenario B (adds €1.2 M avoided reactor and dynamic-support value; residual-value line removed)	+0.90	+0.76	-0.71
Net NPV—Scenario C (design-state life value replaced by €2.40/€2.57 M protection value, Eq. 8)	+0.79	-1.10	-2.43
Net NPV—Scenarios B and C coincident	+3.25	+3.11	+1.79

Note: All figures are indicative; capital costs are planning values, not quotations. NPV = net present value; SVC = static var compensator; STATCOM = static synchronous compensator; TV = tertiary voltage; O&M = operation and maintenance.

Scenario A confirms that replacing a healthy reactor on a thermally normal transformer is not economic, and the hourly integration widens rather than narrows that gap relative to the earlier single-point estimate: the SVC now misses break-even by €1.56 M rather than €0.71 M, because the design-state transformer-life credit has collapsed from €0.90 M to €0.05 M. Simple payback on energy alone remains 21 years for the SVC. Scenario C, which matches the diagnosed unit, carries the SVC to +€0.79 M while the tertiary STATCOM remains €1.10 M short and the 150 kV machine €2.43 M short. Scenario B is positive for both tertiary-connected options but not for the 150 kV machine, whose capital cost is not recovered by an avoided reactor and a dynamic-support credit alone; this is a change from the earlier analysis, which found all options positive in Scenario B. Where the two scenarios coincide—an old reactor on a transformer with a diagnosed hot tertiary, a realistic combination since both stem from the same decades of service—every option is positive and the SVC leads at +€3.25 M.

Sensitivities follow directly from Eq. (2) and the annuity structure. A shift of ±€25/MWh in energy price moves all net present values by €0.41–0.47 M. Each ±1 MVAr in Qrms moves the controllable options by only €0.05–0.06 M, the quadratic suppression again. Each ±10 % of the duty ratio moves the baseline cost, and hence every saving, by €0.26 M. Each ±€1/kVAr·yr of assumed dynamic-support value moves the STATCOM net present values by ±€0.58 M, which makes that single unstated planning assumption the largest lever in Scenario B and the one most in need of replacement by an operator-specific valuation. In Scenario C the fault-progression rate dominates: the deferral term ranges from €0.94 M to €2.25 M across progression rates of 0.5 to 1.5 K/yr, which alone spans €1.3

M of net present value, and the reclamation bias of Section 3.5 can only raise the Scenario C values. The decision is therefore governed by the transformer's actual tertiary condition, the reactor's remaining life and the system valuation of dynamic support—not by the energy price. Table 8 summarises the comparison across all criteria.

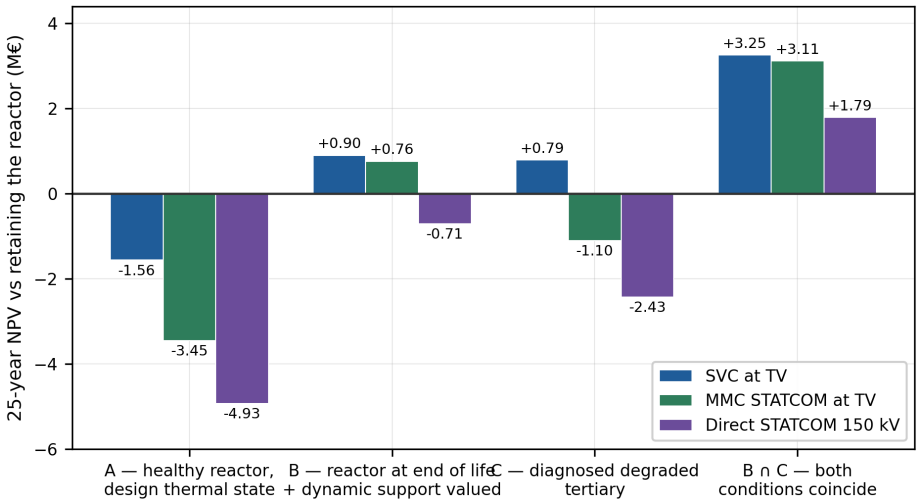


Figure 7. Twenty-five-year net present value against retaining the existing reactor

Note: NPV = net present value; SVC = static var compensator; STATCOM = static synchronous compensator; TV = tertiary voltage.

Table 8. Overall comparison at equal 50 MVAr net inductive rating

Criterion	Reactor at TV	SVC at TV	STATCOM at TV	STATCOM at 150 kV
Rating and control	50 MVAr, fixed block	-12 to +50 MVAr, continuous (TCR rated 62 MVAr)	±50 MVAr, continuous	±50 MVAr, continuous
Average losses/annual cost	254 kW/€222 k	92 kW/€80 k	76 kW/€66 k	71 kW/€62 k
Tertiary loss-of-life rate, design state	0.027 eq-yr/yr	0.0014 eq-yr/yr	0.0014 eq-yr/yr	0.0009 eq-yr/yr
Fault-zone rate, diagnosed unit	≥0.70 eq-yr/yr	0.0014–0.036	0.0014–0.036	0.0014–0.036
Whole-unit ageing	1× (reference)	2.6× slower	2.6× slower	2.8× slower, including events
Works during transformer outage	No	No	No	Yes
Harmonic loss factor at tertiary	1.00 (inrush only)	1.05–1.16	1.00	1.00 (TV unloaded)
Response time	Breaker operation	1–2 cycles	0.25–1 cycle	0.25–1 cycle
Commercial maturity	Mature	Mature	Mature	No transmission-voltage field precedent
Indicative capital cost	Sunk	€3.0 M	€5.0 M	€6.5 M
NPV, Scenarios A/B/C (M€)	0 (reference)	-1.56/+0.90/ +0.79	-3.45/+0.76/ -1.10	-4.93/-0.71/ -2.43

Note: TV = tertiary voltage; SVC = static var compensator; STATCOM = static synchronous compensator; TCR = thyristor-controlled reactor; NPV = net present value; eq-yr/yr = equivalent years per year.

3.9 Limitations

The ambient and through-load series are modelled rather than logged; substituting measured series is straightforward and would remove the largest remaining modelling assumption. A single representative reactive-duty curve is used, although Table 3 shows the loss result is robust to it. Capital costs are planning values. A single diagnosed unit characterises the degraded state, and the furan-to-DP correlations themselves carry roughly ±20% class uncertainty, as the spread of the three DP estimates in Table 5 shows. The polarisation-index and carbon-oxide evidence has been explicitly downgraded in Section 3.4 and the conclusions do not rest on either. The 2026 furan sample post-dates oil reclamation, so the furan-derived quantities are bounds rather than point values. The analysis omits the system-level

cost of the fixed reactor's chronic over-compensation, including tap-changer wear and suboptimal voltage profiles, which would further favour the controllable options; and it omits converter availability and spares strategy, which would work in the opposite direction.

4 Conclusions

For a 280 MVA, 400/150/30 kV OFAF autotransformer whose 50 MVAr tertiary reactor is energised 80% of the year, the statistical loss model and an hourly integration of the IEC 60076-7 loss-of-life integral yield four conclusions.

First, the loss case is strong and robust. Any continuously controlled 50 MVAr compensator cuts annual compensation losses from 2.22 GWh to 0.62–0.80 GWh, worth €142–160 k/yr at Greek price levels, because expected losses depend on the duty only through its mean-square value. The result survives a threefold variation in the assumed Qrms, which is the single least certain input.

Second, the relative thermal benefits claimed for tertiary unloading are confirmed by hourly integration but their absolute magnitude does not support a design-state economic case. The tertiary hot-spot falls 22 K on an annual mean and 28 K during reactor-on hours, the top oil falls 8–11 K, and the whole unit ages 2.6–2.8 times more slowly; yet in the design thermal state the tertiary ages at 0.027 equivalent years per year and the MV winding at 0.010, so the life saved over 25 years is worth under €0.05 M. This corrects a factor-of-eighteen overstatement in the author's earlier single-point treatment, which had evaluated the exponential ageing law at a summer peak-load condition that does not coincide with reactor duty. The correction matters generally: because reactor duty exists precisely because the network is lightly loaded, any tertiary-reactor ageing assessment evaluated at a design point will overstate the benefit of removing it.

Third, the diagnosed unit is governed by a localised fault, not by bulk thermal duty. The furan-derived rate of at least 0.70 equivalent years per year, a lower bound because the sample post-dates oil reclamation, exceeds the modelled bulk rate by a factor of at least 26, and is reproduced only by a fault zone running at least 28 K above the bulk tertiary hot-spot—a mean of 93 °C and a peak of 116 °C at that bound, consistent with the IEC 60599 T1 gas signature and with the 105 °C chain-scission threshold. Removing the tertiary current improves that zone by between 20-fold and 500-fold, and the lower bound holds even if the fault is not itself current-driven, because the exponential ageing law converts the 22 K bulk reduction into a large factor regardless of mechanism. This is a weaker causal claim than asserting that replacement removes the driver, and it is one the evidence can actually support.

Fourth, the harmonic differentiator is real but minor. The delta-connected TCR keeps its largest characteristic order, the third at 13.78% of maximum fundamental, circulating within its own delta; what reaches the tertiary gives a harmonic loss factor of 1.05 to 1.16 at the annual duty point, corresponding to 0.1–0.2 kW of additional eddy loss. Worst-case emission occurs mid-band rather than at rated output, which a rated-point study would miss, but the absolute burden does not justify the weight previously placed on it.

The investment verdict is therefore condition-dependent, and the screening recommendation follows directly: dissolved-gas, furan and dielectric triage of the transformer fleet for tertiary distress—with the stated caveats on polarisation index, carbon-oxide ratios and furan sampling relative to oil reclamation—should drive the replacement sequence. Against a healthy reactor on a thermally normal transformer no option pays back. On a transformer in the diagnosed condition the SVC is positive at +€0.79 M and the STATCOM options are not. When the reactor also reaches end of life and dynamic reactive support is monetised, both tertiary-connected options are positive and every option achieves a positive NPV, from +€1.79 M to +€3.25 M, where the two conditions coincide. Among the technologies, the SVC is the cheapest route to the loss and thermal benefits and its harmonic penalty is now shown to be small; the tertiary STATCOM adds continuous bipolar range, quarter-cycle response and full reactive current at depressed voltage; and the stand-alone 150 kV machine is the only option that removes compensation current from the transformer under all conditions and keeps voltage control available during transformer outages, but it carries the highest capital cost and, on present evidence, no transmission-voltage field precedent for direct connection.

Data Availability

The data used to support the research findings are available from the corresponding author upon request.

Conflicts of Interest

The author declares no conflicts of interest.

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Nomenclature

Symbol/Abbreviation	Description
2FAL	2-furfuraldehyde
DGA	Dissolved gas analysis
DP	Degree of polymerisation of cellulose
EHV	Extra-high voltage
FFT	Fast Fourier transform
HV/MV/TV	High-, medium- and tertiary-voltage winding
MMC/MMCC	Modular multilevel converter/modular multilevel cascade converter
NPV	Net present value
OFAF	Oil forced, air forced cooling
PI	Polarisation index
SDBC/DSCC	Single-delta bridge cells/double-star chopper cells
STATCOM	Static synchronous compensator
SVC	Static var compensator
TCG	Total combustible gas
TCR	Thyristor-controlled reactor
b_{T}	Transformer HV–TV quadratic loss coefficient, kW/MVAr^2
d	Reactor duty ratio
F_{HL}	Harmonic loss factor (IEEE Std C57.110)
g_{r}	Rated winding-to-oil gradient, K
H	Hot-spot factor
h	Harmonic order
K	Load factor, p.u.
L	Loss of life, equivalent years
P_0/P_{k}	No-load/load loss, kW
Q_{rms}	Root-mean-square annual reactive absorption, MVAr
R	Loss ratio (load loss/no-load loss)
V	Relative ageing rate
x, y	Oil and winding exponents of IEC 60076-7
$\Delta\theta_{\mathrm{f}}$	Calibrated fault-zone temperature offset, K
$\Delta\theta_{\mathrm{o}}$/$\Delta\theta_{\mathrm{or}}$	Top-oil rise/rated top-oil rise, K
θ_{a}/θ_{HS}	Ambient/hot-spot temperature, °C